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Review

# Role of Nutrition Therapy in Prediabetes: A Systematic Review

**Hien Thi Nguyen[1], Thang Nguyen[2], Phuc Duong Phan[1], Hai Thanh Phan[3], Huu Minh Le[1], Dat Tan Nguyen[1], Dinh Kim Luong[4], Nam Ba Nguyen[5], and Giang Hai Ha[6,7]**

[1]Faculty of Public Health and [2]Faculty of Pharmacy, Can Tho University of Medicine and Pharmacy, Can Tho 94000, Vietnam
[3]School of Preventive Medicine and Public Health, Hanoi Medical University, Ha Noi 100000, Vietnam
[4]AMV Group Medical Joint Stock Company, Ho Chi Minh City 70000, Vietnam
[5]Jio Health Vietnam General Clinic, Ho Chi Minh City 70000, Vietnam
[6]Institute of Theoretical and Applied Research, Duy Tan University, Ha Noi 100000, Vietnam
[7]School of Business and Economics, Duy Tan University, Da Nang 550000, Vietnam

**ABSTRACT:** Prediabetes represents a critical window for diabetes intervention, but the comparative effectiveness of dietary strategies remains uncertain. This systematic narrative review evaluated structured dietary interventions (SDIs) on metabolic outcomes in adults with prediabetes and identified approaches that may support glycemic improvement and diabetes prevention. PubMed, Embase, the Cochrane Library, and Web of Science were searched for randomized controlled trials (RCTs) published through August 23, 2025. Eligible studies compared SDIs with comparators in adults with prediabetes; observational and nonrandomized trials were excluded. Risk-of-bias (RoB) was assessed using the Cochrane RoB 2 tool. Because of substantial heterogeneity in interventions, comparators, follow-up, and outcomes, findings were synthesized narratively. Finally, ten RCTs, involving 1,098 participants, were included. At 12–52 weeks, calorie-restricted and low-carbohydrate diets were associated with reductions in body weight and glycemic indices, with some trials reporting normoglycemia. Diets for low-glycemic-index appeared to improve insulin sensitivity, whereas Mediterranean diets improved fasting glucose and ectopic fat. High-protein, low-glycemic-index diets may sustain longer-term metabolic enhancements. Intermittent fasting, particularly alternate-day fasting, appeared beneficial for selected outcomes, including weight loss, with glycemic advancements in a few studies. However, interpretation is limited by the small number of trials, heterogeneity, and RoB concerns. Overall, SDIs may offer metabolic benefits for adults with prediabetes, but no single strategy emerged as universally optimal. These findings provide evidence-informed dietary considerations, and further high-quality trials are needed to strengthen sustainable diabetes prevention strategies.

**Keywords:** carbohydrates, glycemic index, insulin, metabolic syndrome, prediabetic state

## INTRODUCTION

Prediabetes, characterized by abnormally high blood glucose levels below the type 2 diabetes (T2D) threshold (Hostalek, 2019), has reached epidemic proportions globally. While prevalence ranges from 3.0% in Southeast Asia to 15.4% in North America (Hostalek, 2019) and diagnostic criteria vary across studies (Lee et al., 2019), there is widespread agreement that diagnoses are rising dramatically (Tabák et al., 2012; Hostalek, 2019). Such a condition is not merely a transitional phase but an independent risk factor for mortality, cardiovascular disease, and microvascular complications (Schlesinger et al., 2022; Ong et al., 2023). Notably, subclinical myocardial necrosis is common even in stable prediabetic patients, signaling a high long-term risk of cardiovascular events (Son et al., 2022). However, prediabetes is highly reversible (Tuso, 2014); comprehensive lifestyle changes can significantly lower the risk of developing T2D (Uusitupa et al., 2019) and cardiovascular disease (Liu et al., 2018). According to the American Diabetes Association (ADA) Standards of Care in Diabetes—2026, early and aggressive individualized medical nutrition therapy is the cornerstone of management (American Diabetes Association Professional Practice Committee for Diabetes, 2026a). Specifically, targeted weight loss of 7%–10% through dietary intervention is extremely effective in halting metabolic deterioration and maximizing normoglycemia reversion (American Diabetes Association Professional Practice Committee for Diabetes, 2026b). Indeed, though a

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Correspondence to Giang Hai Ha, E-mail: hahaigiang@duytan.edu.vn

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healthy diet and physical activity are vital (Galaviz et al., 2022), the role of nutritional therapy in preventing pathogenesis is confirmed (Evert et al., 2019).

Despite a broad consensus on the importance of dietary modification, an optimal nutritional strategy remains intensely debated. Current dietary interventions are highly heterogeneous, ranging from traditional patterns like the traditional Jiangnan diet (Luo et al., 2022), the Mediterranean diet (Huo et al., 2015), Dietary Guidelines for Americans (DeSalvo et al., 2016), to low-carbohydrate approaches (Huntriss et al., 2018) and specific nutritional supplements (Röhling et al., 2020). Furthermore, recent randomized controlled trials (RCTs) have yielded conflicting evidence regarding the comparative efficacy and long-term sustainability of these competing strategies. For instance, while low-carbohydrate and ketogenic diets often yield rapid initial improvements in insulin sensitivity, their long-term impact on ectopic fat accumulation and lipid profiles compared to Mediterranean or low-glycemic index (GI) diets remains inadequately resolved. Systematic reviews have attempted to evaluate these diets but were often constrained by outdated trial data, narrow scopes of study, or a primary focus on short-term glycemic markers rather than the ultimate clinical endpoint of disease reversal (Ge et al., 2020; Goldenberg et al., 2021). This complex and constantly evolving evidence base exacerbates existing systemic barriers in clinical practice. According to the U.S. Centers for Disease Control and Prevention, public awareness of prediabetes remains remarkably low at ~15% (U.S. Centers for Disease Control and Prevention, 2026). Furthermore, profound knowledge gaps persist among both patients and healthcare professionals (HCPs). Only 18.2% of patients recognize that weight loss can reverse prediabetes, while 43% erroneously believe medication is the sole remedy (Somerville et al., 2019), despite official health guidelines advising against drugs or alternative treatments as primary management (Galaviz et al., 2022). Consequently, motivated patients often report frustration over receiving insufficient or inconsistent dietary advice from their HCPs (Teoh et al., 2023), while providers themselves often lack proactivity in referring patients to dietitians (Somerville et al., 2019). Compounding this critical translation gap, the most recent ADA Consensus Report on Nutritional Therapy for prediabetes was published as far back as 2018 and relied solely on PubMed data (Evert et al., 2019).

A comprehensive, up-to-date synthesis is urgent to navigate this complex evidence base. Therefore, this systematic review aims to synthesize the available data from RCTs published up to 2025 evaluating the impacts of diverse dietary interventions on metabolic outcomes in individuals with prediabetes. By critically appraising the methodological quality of recent trials and comparing the outcomes of these strategies, this review seeks to identify specific dietary patterns that may support meaningful metabolic improvements and potential reversion to normoglycemia. Ultimately, this study may provide evidence-informed considerations to assist clinicians, nutritionists, and patients in discussing appropriate dietary approaches tailored to individual needs and preferences.

## METHODS

### Search strategy

This systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 statement (PRISMA 2020) (Page et al., 2021). The protocol was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number: CRD42023458958). The conduct and reporting of the review were consistent with the registered protocol, with no major deviations. The complete PRISMA for Abstracts and PRISMA 2020 checklists are provided in Supplementary Table 1 and 2, respectively. The study selection process and reasons for exclusion are presented in the PRISMA flow diagram in Fig. 1.

Four electronic databases, namely PubMed/MEDLINE, Embase, the Cochrane Library, and Web of Science, were comprehensively searched from database inception to August 23, 2025, without restrictions on publication date or language. The search strategy was developed using the Population, Intervention, Comparator, and Outcomes (PICO) framework to identify RCTs evaluating dietary or nutritional interventions for adults with prediabetes and reporting clinically relevant metabolic outcomes. The detailed search strategies for all electronic databases are provided in Supplementary Table 3.

The search strategy included three main concepts: prediabetes, nutritional therapy, and diabetes progression/metabolic outcomes. Terms within each concept were combined using the Boolean operator "OR," and the three concepts were combined using "AND." Controlled vocabulary terms were utilized where applicable, including MeSH terms in PubMed/MEDLINE and the Cochrane Library, and Emtree terms in Embase. These were combined with free-text terms searched in titles, abstracts, and keywords, per the syntax and indexing structure of each database.

The search strategy included three components:

(1) Prediabetes: Prediabetes OR Pre-Diabetes OR "Intermediate hyperglycemia" OR "Intermediate hyperglycemia" OR "Impaired glucose tolerance" OR Hyperglycemia OR "Glucose Intolerance"
(2) Nutritional therapy: "Dietary guidelines" OR "nutrition guidelines" OR "nutritional therapy"

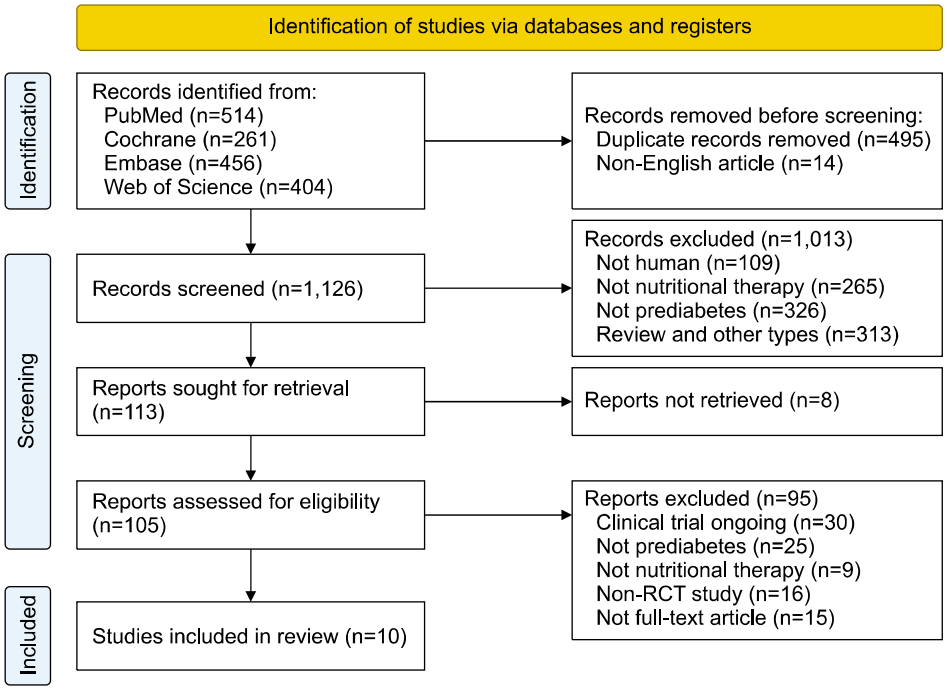



**Fig. 1.** Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram outlining the study selection process. The diagram illustrates the systematic screening, eligibility assessment, and final inclusion of randomized controlled trials (RCTs) evaluating dietary interventions for adults with prediabetes.

OR "nutrition counseling" OR "nutrition education" OR "medical nutrition" OR "diet counseling" OR "healthy eating" OR "healthful eating" OR "eating plan" OR "meal planning" OR "portion control" OR "plate method" OR "carbohydrate counting" OR "calorie counting" OR "mindful eating" OR "intuitive eating" OR "diabetes self-management education" OR "Meal Frequency and Timing" OR "meal timing" OR "Intermittent fasting" OR "USDA Dietary Guidelines For Americans" OR "Mediterranean-style" OR "Vegetarian or vegan" OR "Low-carbohydrate" OR "Very low-carbohydrate" OR "Dietary Approaches to Stop Hypertension" OR "carbohydrate gram counting" OR "Zimbabwean hand jive" OR "Plate method"

(3) Glycemic and metabolic outcomes: "HbA1C" OR "Body weight" OR "MAU" OR "eGFR" OR "Lipid profile" OR "HOMA-IR" OR "QUICKI" OR "Glucose tolerance"

Outcome-related terms were retained as a mandatory component of the search strategy because this review aimed to evaluate the effectiveness of dietary and nutritional interventions on measurable glycemic, anthropometric, insulin-resistance, renal, and lipid-related outcomes in adults with prediabetes. Given the broad and heterogeneous literature on prediabetes, dietary behaviors, lifestyle counseling, and nutrition education, the outcome component was included to improve search specificity and reduce retrieval of studies that discussed dietary or counseling approaches without assessing relevant clinical or metabolic endpoints.

We acknowledge that employing outcome-related terms as mandatory search components may reduce sensitivity, as some eligible studies may not use specific outcome terms in searchable fields such as titles, abstracts, or indexing terms. To minimize this potential limitation, we did not apply any language or publication-date restrictions and manually screened the reference list of all included studies and relevant recent review articles to identify additional eligible trials. This search design has also been acknowledged as a methodological limitation in the discussion section.

To ensure transparency and reproducibility, the complete database-specific search strings for PubMed/MEDLINE, Embase, the Cochrane Library, and Web of Science are provided in Supplementary Table 3. Trial registries and gray literature were not searched.

### Study selection

We structured the eligibility criteria using the PICOS framework (with S denoting study design), following guidance from JBI and the Cochrane Collaboration (Stern et al., 2014). The PICOS criteria applied were:

- P (Population): Adults with prediabetes, as defined by the study authors, including impaired fasting glucose, impaired glucose tolerance, or glycated hemoglobin (HbA1c) within the prediabetic range.
- I (Intervention): Any defined dietary or nutritional intervention; e.g., Mediterranean, low-carbohydrate, vegetarian/vegan, intermittent fasting, portion-control strategies, dietary counseling/education, or structured meal plans.
- C (Comparator): Usual care, minimal intervention, or alternative/active dietary comparator.
- O (Outcomes): Primary outcomes included glycemic control, assessed using HbA1c and fasting glucose,

and insulin resistance, assessed using homeostatic model assessment of insulin resistance (HOMA-IR). Secondary outcomes included anthropometric and metabolic measures, such as body weight/body mass index (BMI), waist circumference (WC), total cholesterol, low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), and triglycerides.

- S (Study design): RCTs only.

All retrieved records were imported into EndNote (version 20, Clarivate), and duplicates were removed via electronic and manual inspection of titles/authors. Screening was managed in Microsoft Excel on Google Drive. Titles and abstracts were independently screened by two reviewers against the PICOS criteria. Full texts of potentially eligible articles were then retrieved and independently assessed by two reviewers for final inclusion. Disagreements at either stage were resolved by discussion; if consensus was not reached, a third senior reviewer adjudicated.

***Exclusion criteria*:** We excluded non-RCT designs and publication types that did not meet our eligibility criteria, specifically: nonhuman, cross-sectional, cohort, or case-control studies, case reports/series, editorials, letters, commentaries, conference abstracts, study protocols, narrative reviews, systematic reviews, and meta-analyses.

### Data extraction and quality assessment

Data were extracted independently by three reviewers and cross-verified by additional ones to ensure accuracy and completeness. Discrepancies were resolved through group discussion until full consensus was achieved.

Extracted data included: (a) article metadata including author, year, title, and country; (b) study characteristics including design, setting, and participants; (c) intervention details including type and description of dietary intervention, baseline measures including HOMA index, body weight, and lipid profile; and (d) outcomes including postintervention HOMA index, weight change, and lipid profile. The primary outcomes were glycemic control, assessed by HbA1c (% or mmol/mol), and the HOMA index. Secondary outcomes included BMI ($kg/m^2$); WC (cm); and lipid parameters including HDL-C, LDL-C, and triglycerides (mmol/L).

The risk-of-bias was assessed using the Cochrane RoB 2 tool, which evaluates five domains. Each study was assigned an overall judgment of "low risk," "some concerns," or "high risk" of bias. Assessments were independently conducted by two reviewers, with disagreements resolved by consensus or consultation with a third reviewer (Sterne et al., 2019).

### Data analysis and synthesis

Due to the substantial clinical and methodological heterogeneity regarding dietary interventions, baseline population characteristics, and outcome measures across the included trials, a quantitative meta-analysis was deemed inappropriate. Consequently, statistical pooling, formal sensitivity analyses, and certainty grading were excluded. Instead, the findings were systematically synthesized narratively in strict accordance with the reporting guidelines of "Synthesis Without Meta-analysis." To ensure absolute transparency and structural rigor, this qualitative synthesis was operationalized through a formalized four-step framework:

***Grouping and definition of interventions*:** The ten RCTs included were categorized into four primary therapeutic dietary models based on their core macronutrient or behavioral characteristics: (i) calorie and carbohydrate restriction, (ii) carbohydrate quality modification (e.g., low- GI diets), (iii) specific holistic dietary patterns (e.g., Mediterranean diet), and (iv) intermittent-fasting regimens.

***Determination of effect direction*:** For each metabolic metric, the direction of effect, beneficial change, worsening, or neutral/no change was determined solely by whether the difference in postintervention outcomes between the active dietary intervention group and its comparator was statistically significant ($P<0.05$).

***Data prioritization*:** Glycemic control markers, specifically HbA1c and fasting blood glucose, were prioritized as primary clinical endpoints to evaluate prediabetes progression or reversion, while anthropometric profiles (body weight, BMI, WC) and lipid fractions were treated as secondary outcomes.

***Integration of risk-of-bias*:** Methodological quality judgments derived from the quality appraisal stage were actively integrated into narrative interpretation and summary tools, ensuring that the strength of clinical conclusions directly reflected the underlying certainty of the evidence.

Before narrative synthesis, data were standardized to address baseline variability and improve comparability across studies. Given the anthropometric differences among the baseline cohorts (e.g., Asian and Western populations), weight change was standardized as the percentage of body weight lost, calculated as follows: mean weight change divided by mean baseline body weight and multiplied by 100. Similarly, biochemical parameters were converted to standard units; for example, glucose and cholesterol fractions were expressed in mg/dL.

Screening, data extraction, and narrative tabulation were managed using Microsoft Excel via Google Drive, with structured summary tables and visual tools generated to ensure a transparent presentation of effect directions and magnitudes. Methodological quality was evaluated employing the RoB 2 domain-based tool for randomized trials (Cochrane Methods Bias, 2019). This framework applies standardized signaling questions within five distinct bias domains to yield algorithm-based judgments

categorized as "low risk," "some concerns," or "high risk" of bias. To maximize reporting transparency while accommodating manuscript presentation-related constraints, overall risk-of-bias summaries were systematically embedded within the structured outcome summary tables alongside the corresponding findings.

## RESULTS

For reports whose full texts were not immediately accessible, requests were made through interlibrary loan services, and the authors were contacted via email. Studies were excluded only after these attempts failed to yield the data necessary for quality assessment. Fig. 1 presents the selection process. After removing duplicates, a total of 1,126 articles were screened per the eligibility criteria, and 105 publications were included for full-text review; ten papers met the inclusion criteria and were included in the final analysis.

### Quality assessment

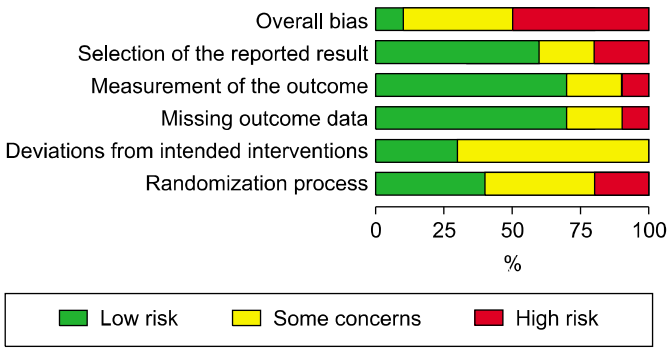
Risk of bias was assessed using the RoB 2 tool (Fig. 2). For selective reporting, 60% of studies were judged to be at low risk, 20% to have some concerns, and 20% to be at high risk. Outcome measurement was generally robust, with 80% of studies judged to be at low risk. Missing outcome data were also generally well handled, with 70% of studies judged to be at low risk, 20% to have some concerns, and 10% to be at high risk. In contrast, deviations from intended interventions frequently raised concerns: 70% of studies were judged to have some concerns and 30% to be at low risk. The randomization process also showed notable inconsistencies, with 40% of studies judged to be at low risk, 40% to have some concerns, and 20% to be at high risk.

Overall, five studies (50%) were rated with high risk-of-bias, four (40%) raised some concerns, and only one (10%) was considered low risk. Consistent with Cochrane and PRISMA guidelines, no study was excluded solely due to high risk for a single domain. Instead, all ten studies were retained to maximize comprehensiveness, with identified risks carefully integrated into a cautious narrative interpretation of the findings.

### Study characteristics

Ten studies involving both community- and clinic-based populations from diverse geographic settings, including China, Germany, Austria, the United Kingdom, France, the United States, Australia, Canada, Korea, Mexico, and Iran, were included (Table 1). Intervention durations ranged from 3 weeks to 3 years. Participants were adults aged 18 – 65 years, and most studies reported mean participant ages above 55 years.

Based on their nutritional characteristics, the 18 dietary interventions were grouped into four categories: (i) calorie- and carbohydrate-restricted diets (Diets 1, 6, and 12), which emphasized reduced energy and carbohydrate intake with variations in protein and unsaturated fat; (ii) carbohydrate quality-focused approaches (Diets 8, 9, 11, 13, and 14), which prioritized low-glycemic-index foods, whole grains, and legumes while limiting refined carbohydrates; (iii) specific dietary patterns (Diets 2, 3, 10, 15, 17, and 18), including plant-forward, high-fi-




| Study | D1 | D2 | D3 | D4 | D5 | Overall |
|---|---|---|---|---|---|---|
| Röhling et al., 2020 | ! | ! | ! | ! | ! | ! |
| Chair et al., 2022 | + | ! | + | − | + | − |
| Buso et al., 2021 | ! | + | + | + | − | − |
| Bartholomew et al., 2021 | + | ! | ! | + | + | ! |
| Wolever and Mehling, 2003 | − | ! | + | + | + | − |
| Kim et al., 2014 | ! | ! | − | + | − | − |
| Velázquez-López et al., 2013 | ! | ! | + | + | + | ! |
| Wolever and Mehling, 2002 | − | ! | + | + | + | − |
| Luo et al., 2022 | + | + | + | ! | ! | ! |
| Pasdar et al., 2025 | + | + | + | + | + | + |

D1 Randomization process
D2 Deviations from the intended interventions
D3 Missing outcome data
D4 Measurement of the outcome
D5 Selection of the reported result

+ Low risk
! Some concerns
− High risk

**Fig. 2.** Methodological quality and risk of bias assessment using the Cochrane risk of bias 2.0 (RoB 2) tool. Visual summary of bias risk across five domains: (D1) bias arising from the randomization process, (D2) bias due to deviations from intended interventions, (D3) bias due to missing outcome data, (D4) bias in measurement of the outcome, and (D5) bias in selection of the reported result.

**Table 1.** Characteristics and main findings of the included studies

| Study | Participant (n, country) | Age (years) | Duration (weeks) | Dietary or nutritional intervention | Effectively controls blood sugar and improves insulin resistance | Body composition changes (weight change: kg or %) | Effectively regulates blood lipids profile |
|---|---|---|---|---|---|---|---|
| Röhling et al., 2020 | 141 (Germany, Austria, UK, France) | 21–65 | 52 | (1) Low-calorie and low-carb diet<br>Week 1: 3 meals powder<br>Weeks 2–4: 2 meals powder+low-carb lunch<br>Weeks 5–26: 1,300–1,500 kcal/d diet<br>After 26 weeks: received a cookbook with low-carb and healthy meal ideas | (1) After 12 weeks:<br>↓HbA1C*: ↓0.19%<br>After 52 weeks:<br>↓HbA1C**, 40% from prediabetes to normoglycemia (HbA1c<5.7%) | (1) After 12 weeks:<br>Weight***: ↓5.6 (6%)<br>BMI***: ↓1.9 (5.9%)<br>WC**: ↓4.8<br>FM***: ↓4.5<br>FFM***: ↓1.0<br>After 52 weeks:<br>Weight**: ↓4.1<br>BMI**: ↓1.4<br>FM*: ↓3.2 | (1) After 12 weeks:<br>TC*: ↓0.41<br>LDL-C**: ↓0.34 |
| Luo et al., 2022 | 253 (China) | 25–60 | 24 | (2) MD: Emphasizes olive oil, marine fish, nuts, and whole grains; dishes are prepared at lower temperatures to suit Chinese preferences<br>(3) TJD: Typical Southeast Chinese cuisine, rich in vegetables, fruits, legumes, soy, freshwater fish, and shrimp, with moderate whole grain intake (e.g., brown rice)<br>CD: High in saturated fat, pork, and refined rice, reflecting common urban dietary patterns (e.g., Shanghai) | (2) MD<br>3 months:<br>FBG***: ↓0.12<br>6 months:<br>FBG***: ↓0.38<br>(3) TJD<br>3 months:<br>FBG***: ↓0.14<br>6 months:<br>FBG***: ↓0.34<br>CD<br>3 months:<br>FBG***: ↓0.12<br>6 months:<br>FBG***: ↓0.28 | (2) MD: Average weight loss at 6 months: ↓5.72 (~7%)***<br>(3)<br>TJD: Weight*** ↓5.05<br>CD: Weight*** ↓5.38 | Not significant |
| Chair et al., 2022 | 101 (China) | 18–65 | 12 | (4) ADF: 600 kcal on fasting days, alternating with ad libitum intake on eating days<br>(5) 16/8 TRE: All food consumed within an 8-h daily window, followed by a 16-h fast; participants chose the 8-h period that fit their routine<br>T1 (end of 3 weeks intervention), T2 (3-month follow-up) | (4) ADF<br>T1: FBG* ↓0.13 ($P$=0.027)<br>T2: FBG* ↓0.13 ($P$=0.020)<br>(5) 16/8 TRE<br>T1: FBG* ↓0.16 ($P$=0.014)<br>T2: FBG*** ↓0.25<br>Alternate model (T2):<br>FBG** ↓0.19 | (4) ADF<br>T1***: Weight ↓14.72; BMI ↓3.69; WC ↓8.16<br>T2***: Weight ↓14.80; BMI ↓3.69; WC ↓8.34<br>(5) TRE<br>T1***: Weight ↓4.55; BMI ↓1.59; WC ↓5.54<br>T2***: Weight ↓4.44; BMI ↓1.55; WC ↓5.67 | (4) ADF<br>TC*: ↓0.79 at T1 vs. control<br>TC*: ↓0.80 vs. TRE<br>TG*: ↑0.32 at T1 vs. TRE<br>TG**: ↑0.46 at T2 vs. TRE<br>(5) TRE<br>TG***: ↓at T1 & T2 (vs. baseline)<br>TG*: ↓0.34 (vs. control) at T1<br>TG***: ↓0.66 (vs. control) at T2 |

**Table 1.** Continued 1

| Study | Participant (n, country) | Age (years) | Duration (weeks) | Dietary or nutritional intervention | Effectively controls blood sugar and improves insulin resistance | Body composition changes (Weight change: kg or %) | Effectively regulates blood lipids profile |
|---|---|---|---|---|---|---|---|
| Buso et al., 2021 | 136 (Australia) | 48–62 | 156 | (6) Higher-protein, low-GI diet<br>Phase 1 (2-month weight loss): 800 kcal/d with defined proportions of fat, protein, and carb<br>Phase 2 (36-month maintenance):<br>HP/LGI: Higher-protein, moderate-carb, low-GI diet<br>MP/MGI: Moderate-protein, higher-carb, moderate GI diet<br>Both consumed *ad libitum* (no calorie restriction) | (6)<br>Phase 1*: 40.2% Achieved NG<br>Phase 2*: 18.2% Sustained NG. Improvements observed for HOMA-IR, FI, and 2-h glucose (all $P<0.05$) | (6)<br>Phase 1***: 100% Achieved ≥8% weight loss (median ~11.0%) vs. baseline<br>Phase 2***: 48.7% Of lost weight was regained, yet weight remained significantly below baseline ($P<0.001$) | NA |
| Bartholomew et al., 2021 | 103 (USA) | 21–70 | 26 | (7) IF24: Participants fasted twice weekly for 24 h (21–27 h) during the first 4 weeks, then once weekly for 24 h over the following 22 weeks | (7) IF24<br>FBG* ↓0.4;<br>FI*** ↓2.76 mIU/L;<br>HOMA-IR improved by 32.5%; MSS** ↓ | Not significant | Not significant |
| Wolever and Mehling, 2003 | 44 (Canada) | 30–65 | 16 | (8) High-carbohydrate, high-GI: 55% Energy from carb, 30% from fat [refined, high GI foods (white rice, potatoes)]<br>(9) High-carbohydrate, low-GI: Same macronutrient ratio, but with low-GI carb, whole grain, low-GI foods (whole grains, beans)<br>(10) Low-carbohydrate, MUFA: 45% carb, 40% fat, with half of fat as MUFAs. Lower carbs, increase good fats (olive oil)<br>At baseline, the mean energy intake of all IGT participants was 7,390 kJ/d (~1,766 kcal/d). During the 4-month intervention, energy intake remained largely unchanged across all three diet groups | (8) High-GI<br>0–8-h Insulin* ↓35.2 pmol/L (vs. baseline)<br>(9) Low-GI<br>PPG* ↓0.35 (vs. high-GI)<br>(10) MUFA<br>PPG* ↓0.35 (vs. high-GI)<br>HbA1c**: ↑0.02%<br>0–8-h Insulin*: ↓28.3 pmol/L (vs. baseline) | Not significant | (8) High-GI<br>TG*: ↓0.14<br>(9) Low-GI<br>TG**: ↑0.18 |
| Kim et al., 2014 | 99 (Korea) | 55.4±1.5 | 12 | (11) Choosing carb sources<br>Replaced refined rice with a mix of black soybeans, barley, and brown rice (1/3 each), and increased vegetable intake for dietary fiber | (11)<br>Glucose***: ↓0.60<br>HbA1c (%)***: ↓0.33±0.12<br>HOMA-IR***: ↓0.44±0.08 | Not significant | Not significant |

**Table 1.** Continued 2

| Study | Participant (n, country) | Age (years) | Duration (weeks) | Dietary or nutritional intervention | Effectively controls blood sugar and improves insulin resistance | Body composition changes (Weight change: kg or %) | Effectively regulates blood lipids profile |
|---|---|---|---|---|---|---|---|
| Velázquez-López et al., 2013 | 86 (Mexico) | 49±10.04 | 52 | (12) Low calorie and carbohydrate diet: Women (1,200 – 1,400 kcal), men (1,500 – 1,700 kcal). The diet included 50% carb, 20% protein, and ≤30% fat, with a fiber intake goal of 25 g per day | (12)<br>After 6 months<br>Glucose***: ↓0.30<br>After 12 months<br>Glucose**: ↓0.57 | (12)<br>Overweight group (n=43):<br>Weight (6/12 months)***: ↓2.5/↓2.8<br>BMI (6/12 months)***: ↓0.9/↓1.0<br>WC (6/12 months)***: ↓2.8/↓3.3<br>HC (6/12 months) (*/**): ↓1.5/↓1.5<br>Fat % (6/12 months) (**/***): ↓2.1/↓2.6<br>FM (6/12 months) (**/***): ↓2.4/↓2.7<br>Obesity group (n=43):<br>Weight (6/12 months)***: ↓3.3/↓5.4<br>BMI (6/12 months)***: ↓1.3/↓2.1<br>WC (6/12 months)***: ↓5.0/↓6.9<br>HC (6/12 months) (*/***): ↓2.1/↓2.8<br>Fat %: (6/12 months)***: ↓1.6/↓2.3<br>FM (6/12 months)***: ↓2.4/↓2.8 | (12)<br>Overweight group:<br>TG (6/12 months) (*/**): ↓0.38/↓0.70<br>TG (6/12 months) (**/***): ↓0.21/↓0.32<br>HDL-C (6/12 months) (**/***): ↑0.13/↑0.14<br>LDL-C (6/12 months) (**/***): ↓0.41/↓0.64<br>Obesity group:<br>TG (6/12 months) (*/***): ↓0.29/↓0.61<br>TC (6/12 months) (**/***): ↓0.28/↓0.65<br>HDL-C (6/12 months) (**/**): ↑0.09/↑0.12<br>LDL-C (6/12 months) (*/***): ↓0.24/↓0.49 |
| Wolever and Mehling, 2002 | 37 Randomized; 34 completed (Canada) | 30 – 65 | 16 | (13) High-carb, high-GI: 53% carb, 28% fat, GI 83<br>(14) High-carb, low-GI: 55% carb, 25% fat, GI 76<br>(15) Low-carb, high-MUFA: 47% carb, 35% fat, GI 82 | (14) The glucose DI* ↑>50%<br>(15) HbA1c: ↑Significantly vs. (13) and (14)<br>(13) & (15): No significant change in DI | Not significant<br>(13) Weight ↓0.49<br>(14) Weight ↓0.19<br>(15) Weight ↑0.27 | Not significant |

**Table 1.** Continued 3

| Study | Participant (n, country) | Age (years) | Duration (weeks) | Dietary or nutritional intervention | Effectively controls blood sugar and improves insulin resistance | Body composition changes (Weight change: kg or %) | Effectively regulates blood lipids profile |
|---|---|---|---|---|---|---|---|
| Pasdar et al., 2025 | 98 Randomized; 83 completed (Iran) | 18–55 | 12 | (16) IF 5:2: Normal eating 5 days; 2 nonconsecutive days with ~25% calories (~500–600 kcal/d)<br>(17) PMC: Plant-based portfolio diet: ~40% carb/20% protein/40% fat; emphasizes plant protein (~50 g), 20 g soluble fiber, ~45 g nuts, ~2 g phytosterols, and ~45 g MUFA/d<br>(18) AI: ~25% calorie restriction+ anti-inflammatory foods (plenty of vegetables, fatty fish, MUFAs); limit refined carbs, saturated/omega-6 fats<br>Control (healthy calorie-restricted): ~25% calorie restriction with standard macros: 50% carb/20% protein/30% fat | (16) IF 5:2<br>HbA1c*** ↓(within group)<br>(17) PMC<br>FBG* ↓(within-group)<br>HbA1c*** ↓(within-group)<br>(18) AI<br>FBG* ↓0.36 (vs. control)<br>HbA1c* ↓0.27% (vs. control)<br>Control<br>FBG* ↓(within-group)<br>HbA1c*** ↓(within-group) | (16) IF 5:2<br>Weight*** ↓ (within-group)<br>BMI* ↓0.33 (vs. control)<br>BF%** ↓1.32% ($P$<0.01)<br>(17) PMC<br>Weight*** ↓ (within-group)<br>BF%** ↓1.53%<br>FFM* ↑2% (vs. control)<br>(18) AI<br>Weight*** ↓1.02 (vs. control)<br>BMI* ↓0.40 (vs. control)<br>BFM** ↓2.27%<br>FFM* ↑(vs. control)<br>Control:<br>Weight*** ↓ (within-group) | (16) IF 5:2<br>TC* ↓(within-group)<br>HDL*** ↑(within-group)<br>(17) PMC<br>Lipids: TC** ↓0.49 (vs. control)<br>LDL** ↓(within-group)<br>hs-CRP** ↓($P$=0.003)<br>(18) AI<br>hs-CRP*** ↓($P$=0.001) |

Study groups and diets: MD, Mediterranean diet; TJD, traditional Jiangnan diet; CD, control diet; ADF, alternate-day fasting; TRE, time-restricted eating; TRF, time-restricted fasting; IF24, 24-h intermittent fasting; IF 5:2, 5:2 intermittent fasting; PMC, portfolio moderate-carbohydrate; AI, anti-inflammatory diet; HP/LGI, high-protein, low-glycemic index; MP/MGI, moderate-protein, moderate-glycemic index; MUFA, monounsaturated fatty acids.
Anthropometrics: BMI, body mass index (kg/m$^2$); WC, waist circumference (cm); HC, hip circumference (cm); FM, fat mass (kg); FFM, fat-free mass (kg); BFM, body fat mass (kg); MSS, metabolic syndrome score.
Blood and metabolic indices: FBG, fasting blood glucose (mmol/L); PPG, postprandial glucose (mmol/L); FI, fasting insulin (μIU/mL); NG, normoglycemia; IGT, impaired glucose tolerance; HbA1c, glycated hemoglobin (%); HOMA-IR, homeostatic model assessment of insulin resistance; TC, total cholesterol (mmol/L); TG, triglycerides (mmol/L); LDL-C, low-density lipoprotein cholesterol (mmol/L); HDL-C, high-density lipoprotein cholesterol (mmol/L); hs-CRP, high-sensitivity C-reactive protein.
Nutrients and others: GI, glycemic index; NA, not available/not applicable.
Symbols: ↑, increase; ↓, decrease.
Only statistically significant between-group changes are shown. Statistical significance is denoted as follows: *$P \leq 0.05$; **$P \leq 0.01$; and ***$P \leq 0.001$.
Total sample size (n=1,098) was calculated based on the number of participants randomized across the included studies.

ber, and anti-inflammatory diets rich in monounsaturated fatty acids, phytosterols, and other bioactive compounds; and (iv) intermittent fasting regimens (Diets 4, 5, 7, and 16), characterized by strategies such as alternate-day fasting (ADF), 24-h fasting, the 5:2 diet, and 16:8 time-restricted eating (TRE).

Control conditions varied substantially across studies, reflecting differences in study design and ethical considerations. Some trials employed structured dietary controls, such as a calorie-restricted healthy diet with standardized macronutrient distribution (Pasdar et al., 2025), or provided general lifestyle advice without specific targets (Röhling et al., 2020). Others maintained habitual diets reflective of local patterns, often rich in refined carbohydrates and saturated fats. Attention-control designs with equivalent educational input but no dietary prescription (Chair et al., 2022), as well as *ad libitum* free-living controls (Bartholomew et al., 2021), were also analyzed. Several studies did not include a conventional non-intervention control group; instead, they relied on participant-participant comparisons or active dietary comparators, with baseline measures as the reference (Wolever and Mehling, 2002, 2003; Velázquez-López et al., 2013; Buso et al., 2021).

Overall, the heterogeneity in control conditions highlights methodological variability across studies and underscores the need for cautious interpretation when comparing intervention effects.

Dietary interventions for prediabetes were associated with variability in body weight, glycemic control, insulin resistance, lipid profiles, and inflammatory markers, with statistically significant clinical outcomes and temporal patterns among approaches systematically aggregated in Table 2.

**Calorie- and carbohydrate-restricted diets (nutritional intervention numbers: 1, 6, and 12)**

These regimens appeared to induce more rapid short-term reductions in weight and glycemic parameters compared to the other evaluated patterns. Short-term weight and fat loss were substantial (e.g., −5.6 and −4.5 kg at 12 weeks with Diet 1), accompanied by modest HbA1c declines and early lipid improvements, though certain attenuation occurred by 52 weeks. Very low-calorie, low-carbohydrate interventions achieved marked early weight loss (~11% at 2 months) and notable reversion to normoglycemia (~40%), with partial long-term maintenance despite weight regain. Longer-term carbohydrate restriction yielded more gradual but durable impacts, including sustained fasting glucose reduction and marked improvements in triglycerides, total cholesterol, LDL-C, and HDL-C at 12 months (Velázquez-López et al., 2013; Röhling et al., 2020; Buso et al., 2021).

**Carbohydrate quality-focused approaches (nutritional intervention numbers: 8, 9, 11, 13, and 14)**

Effects varied by GI and food sources. High-GI diets caused minimal metabolic benefits beyond modest HDL and postprandial insulin changes, with no improvement in HbA1c or insulin sensitivity. In contrast, low-GI, fiber-rich diets enhanced β-cell function, postprandial glycemia, and blood pressure, albeit with mixed lipid responses. A whole-grain-legume-vegetable pattern (Diet 11) consistently reduced fasting glucose, insulin, HbA1c, HOMA-IR, and inflammatory markers, independent of weight loss (Wolever and Mehling, 2002, 2003; Kim et al., 2014).

**Specific dietary patterns: pattern-based, anti-inflammatory, and high-fiber (nutritional interventions 2, 3, 10, 15, 17, and 18)**

Mediterranean and traditional Jiangnan diets achieved ~7% weight loss over 6 months with parallel decreases in fasting glucose and additional benefits in ectopic fat reduction. Portfolio-moderate-carbohydrate and anti-inflammatory diets improved glycemia, body composition, lipid profiles, and high-sensitivity C-reactive protein (hs-CRP), with the anti-inflammatory approach yielding the most pronounced metabolic and inflammatory improvement. Conversely, a low-carbohydrate, high-monounsaturated fatty acid-rich (MUFA) diet increased HbA1c and postprandial triglycerides but reduced β-cell function, suggesting potential metabolic trade-offs (Wolever and Mehling, 2002, 2003; Luo et al., 2022; Pasdar et al., 2025).

**Intermittent-fasting regimens (nutritional intervention numbers 4, 5, 7, and 16)**

Outcomes differed by regimen; ADF induced the greatest weight loss (up to −14.7 kg) and improved fasting glucose but had limited or adverse effects on triglyceride levels. TRE produced moderate weight loss with more favorable lipid and glycemic impacts. Weekly 24-h fasting enhanced insulin sensitivity and metabolic syndrome components without marked changes in weight, while the 5:2 regimen reduced adiposity, fasting glucose, and HbA1c but induced variable lipid and inflammatory responses with moderate adherence challenges (Bartholomew et al., 2021; Chair et al., 2022; Pasdar et al., 2025).

Overall, (i) calorie/carbohydrate restriction was often linked to prominent short-term weight and glycemic reductions; (ii) carbohydrate quality and pattern-based diets appeared to support broader cardiometabolic markers; and (iii) intermittent fasting showed variable efficacy. These findings support dietary strategies tailored to specific clinical goals and patient adherence capacity.

**Table 2.** Statistically significant clinical outcomes of dietary interventions in prediabetic individuals

| The dietary intervention category | Statistically significant clinical outcome | | |
|---|---|---|---|
| | Body composition | Blood glucose and insulin resistance | Blood lipid |
| Calorie- and carbohydrate-restricted diets (nutritional intervention number: 1, 6, 12) | Diet 1: Faster weight and fat loss in 12 weeks (weight ↓5.6 kg, fat mass ↓4.5 kg), with slight FFM loss (−1.0 kg)<br>Diet 6: Rapid short-term weight loss (median ↓11% in 2 months); partial regain over 36 months (48.7%), but weight remained below baseline ($P$<0.001)<br>Diet 12: Slower but sustained weight loss over 12 months (weight ↓2.8 kg), with stable reductions in waist (−3.3 cm), hip (−1.5 cm), and body fat % (−2.6%) | Diet 1: Rapid improvement in HbA1c after 12 weeks (↓0.19%, $P$=0.048); sustained at 52 weeks ($P$=0.008), with 40% of participants reverting from prediabetes to normoglycemia (HbA1c <5.7%)<br>Diet 12: Slower but continuous reduction in glucose over 12 months (6 months: ↓0.30 mmol/L, $P$=0.001; 12 months: ↓0.57 mmol/L, $P$=0.01)<br>Diet 6: Achieved strong short-term reversal of prediabetes (40.2% at 2 months) with partial maintenance at 3 years (18.2% at 3 years, HOMA-IR, insulin improved, $P$<0.05) | Diet 1:Rapid short-term improvements in lipids (12 weeks): total cholesterol ↓0.41 mmol/L, LDL-C ↓0.34 mmol/L<br>Diet 12: Sustained long-term improvements (12 months): triglycerides ↓0.70 mmol/L, total cholesterol ↓0.74 mmol/L, LDL-C ↓0.64 mmol/L, HDL-C ↑0.14 mmol/L |
| Dietary approach focusing on carbohydrate source selection (nutritional intervention number: 8, 9, 11, 13, 14) | Not significant | Diet 11: Most effective; glucose ↓0.60 mmol/L, HbA1c ↓0.33%, HOMA-IR ↓0.44 ($P$<0.001)<br>Diet 14: Moderate improvement in β-cell function (>50% increase in glucose disposition index, $P$=0.02)<br>Diet 8: 0–8 h Insulin ↓35.2 pmol/L vs. baseline<br>Diet 9: PPG ↓0.35 mmol/L vs. high-GI; overall least effective<br>Diet 13: NA | These dietary interventions showed no clear effectiveness on lipid profile |
| Specific dietary patterns (pattern-based, anti-inflammatory, high-fiber) (included nutritional intervention number 2, 3, 10, 15, 17, 18) | Diet 2 (MD): Most effective, ~7% weight loss at 6 months (↓5.72 kg, $P$<0.001)<br>Diet 3 (TJD, CD): Comparable results, weight loss ↓5.05–5.38 kg ($P$<0.001)<br>Diet 17 (PMC) and Diet 18 (AI): significant within-group weight reductions ($P$<0.001), with favorable body composition changes | Diet 2 (MD): Fasting glucose ↓0.12 mmol/L at 3 months, ↓0.38 mmol/L at 6 months ($P$<0.001)<br>Diet 3 (TJD): Fasting glucose ↓0.14 mmol/L at 3 months, ↓0.34 mmol/L at 6 months ($P$<0.001)<br>Diet 18 (AI): Fasting glucose ↓0.36 mmol/L, HbA1c ↓0.27% ($P$<0.05)<br>Diet 17 (PMC): Fasting glucose and HbA1c improved ($P$<0.05)<br>Diet 10, Diet 15 (MUFA): Minimal effect | Diet 17 (PMC): Improved lipid profile with total cholesterol ↓0.49 mmol/L and reduced inflammation (hs-CRP ↓, $P$=0.003)<br>Diet 18 (AI): AI mainly reduced inflammation, hs-CRP markedly decreased ($P$=0.001) |
| Intermittent-fasting regimens (nutritional intervention number: 4, 5, 7, 16) | Diet 4 (ADF): Most effective, weight ↓14.7–14.8 kg, BMI ↓3.69, WC ↓8.2–8.3 cm ($P$<0.001).<br>Diet 5 (TRE): Moderate effect, weight ↓4.4–4.6 kg, BMI ↓1.55–1.59, WC ↓5.5–5.7 cm ($P$<0.001)<br>Diet 16 (IF 5:2): Small effect | Diet 7 (IF24): FBG ↓0.40 mmol/L<br>Diet 5 (TRE): FBG ↓0.16–0.25 mmol/L<br>Diet 4 (ADF): FBG ↓0.13 mmol/L<br>Diet 16 (IF 5:2): Modest effect, mainly HbA1c | Diet 5 (TRE): Lowers triglycerides (↓0.34–0.66 mmol/L)<br>Diet 4 (ADF): Reduces total cholesterol (↓0.79–0.80 mmol/L) but may raise triglycerides |

FFM, fat-free mass; HbA1c, glycated hemoglobin; HOMA-IR, homeostatic model assessment of insulin resistance; LDL-C, low-density lipoprotein cholesterol; HDL-C, high-density lipoprotein cholesterol; PPG, postprandial glucose; GI, glycemic index; MD, Mediterranean diet; TJD, traditional Jiangnan diet; CD, control diet; PMC, portfolio moderate-carbohydrate; AI, anti-inflammatory diet; MUFA, monounsaturated fatty acids; hs-CRP, high-sensitivity c-reactive protein; ADF, alternate-day fasting; BMI, body mass index; WC, waist circumference; TRE, time-restricted eating; IF 5:2, 5:2 intermittent fasting; IF24, 24-h intermittent fasting; FBG: fasting blood glucose.
Only outcomes with $P$<0.05 are reported here.

## DISCUSSION

### Comparative benefits and limitations of dietary patterns

The dietary interventions reviewed here demonstrated distinct benefits and limitations, indicating that no single dietary pattern is universally optimal for all individuals with prediabetes. Rather, the relative effectiveness of each approach appears to depend on the targeted outcome, intervention duration, metabolic phenotype, and long-term adherence.

Calorie- and carbohydrate-restricted diets, particularly low- and very-low-carbohydrate-based approaches, ap-

peared to produce the most pronounced short- to medium-term influence on weight loss and glycemic improvement. These interventions have been associated with rapid reductions in body weight, fasting glucose, and HbA1c, with some trials reporting a substantial reversion from prediabetes to normoglycemia. Formula-based meal replacement strategies may further enhance short-term adherence by simplifying dietary choices and reducing decision burden. However, the durability of these effects remains uncertain. Several studies have reported an attenuation of benefits after one year, partial weight regain during longer follow-up, and a temporally diminishing statistical advantage over control diets. In addition, carbohydrate restriction may induce unfavorable lipid responses in some individuals, including increases in LDL-C or other atherogenic lipid fractions, warranting careful monitoring in those at elevated cardiovascular risk (Velázquez-López et al., 2013; Röhling et al., 2020; Buso et al., 2021).

Evidence from trials in individuals with T2D provides indirect support for the metabolic potential of low-carbohydrate diets. Such interventions have been associated with reductions in HbA1c, body weight, BMI, and WC at 3 – 6 months (Gram-Kampmann et al., 2022; Ahmed et al., 2025). A similar review of 23 trials involving 1,357 participants found that low-carbohydrate diets increased diabetes remission at 6 months, particularly among participants not using insulin, while also improving body weight, triglycerides, and insulin sensitivity. However, these benefits were less consistent at 12 months (Goldenberg et al., 2021). These findings support the biological plausibility of carbohydrate restriction as a short-term metabolic strategy, although extrapolation to prediabetic populations should be cautious.

Dietary approaches emphasizing carbohydrate quality, including low-glycemic-index foods, whole grains, legumes, and other fiber-rich carbohydrate sources, appear to improve postprandial glycemia and insulin sensitivity indices even without substantial weight loss. Improvements in β-cell function, including increases in the disposition index, suggest that replacing refined carbohydrates with fiber-rich, low-glycemic alternatives may enhance glycemic regulation through mechanisms beyond energy restriction alone. These diets may also exert anti-inflammatory and antioxidant effects, as reflected by declines in biomarkers such as interleukin-6, oxidized LDL, and malondialdehyde (Wolever and Mehling, 2002, 2003; Kim et al., 2014).

A key limitation of this approach is that its influence on body weight is generally more modest than that observed with energy- or carbohydrate-restricted diets. Moreover, a few studies have reported mixed lipid responses, including reductions in HDL-C or increases in postprandial triglycerides, although these changes may not necessarily reflect the same pathophysiological processes as insulin resistance-related dyslipidemia. Overall, carbohydrate-quality-focused interventions may be particularly suitable as sustainable long-term strategies for improving glycemic stability and cardiometabolic health, especially in populations where refined grains are staple foods. Evidence from low-glycemic-load and whole-grain dietary patterns further supports potential benefits in reducing systemic inflammation and cardiovascular risk factors among individuals with poor glycemic control (Qi and Hu, 2007).

Specific whole-diet patterns, such as the Mediterranean, traditional Jiangnan, and anti-inflammatory diets, offer wider cardiometabolic benefits. These patterns have been associated with improvements in glycemic control, lipid profiles, inflammation, and overall diet quality. Their strengths lie in high nutrient density and a favorable food matrix, characterized by greater intakes of fiber, monounsaturated and polyunsaturated fatty acids, antioxidants, and bioactive phytocompounds. High adherence rates reported in some studies suggest that culturally familiar and palatable dietary patterns may be more feasible for long-term implementation than highly restrictive regimens (Wolever and Mehling, 2002, 2003; Luo et al., 2022; Pasdar et al., 2025).

Nevertheless, when total energy intake is balanced across intervention groups, differences between dietary patterns often become smaller, suggesting that energy deficit may be a major driver of short-term metabolic improvement. Thus, the specific structure of a diet may be most crucial for long-term adherence, nutritional adequacy, cardiovascular risk reduction, and inflammatory regulation, rather than for glycemic improvement alone. In particular, the Mediterranean diet may support glucose homeostasis through improved insulin sensitivity, favorable modulation of the gut microbiota, and anti-inflammatory and antioxidant pathways (Milenkovic et al., 2021). However, cultural adaptability, food availability, and cost must be considered when translating these dietary patterns to diverse community settings.

Intermittent fasting regimens, including ADF and TRE, potentially benefit body weight, BMI, insulin resistance, and lipid metabolism. ADF appears to more pronouncedly reduce body weight and BMI, whereas 16:8 TRE has favorably affected triglycerides and LDL-C in certain studies. These interventions may improve insulin sensitivity through mechanisms related to metabolic switching, reduced energy intake, circadian alignment, and possibly autophagy, although the independent contributions of these mechanisms beyond weight loss remain difficult to isolate (Bartholomew et al., 2021; Chair et al., 2022; Pasdar et al., 2025).

Despite these potential advantages, intermittent fasting has vital limitations. Dropout rates have been rela-

tively high in a few studies, and adverse symptoms such as hunger, headache, dizziness, and hypoglycemia may compromise adherence and safety, particularly during prolonged fasting. Lipid responses are also inconsistent, with specific evidence suggesting possible elevations in triglycerides under ADF protocols. Therefore, intermittent fasting must not be considered a universally appropriate intervention for individuals with prediabetes, especially those using glucose-lowering medications or with complex comorbidities. Reviews suggest that intermittent fasting may reduce body weight, fasting glucose, insulin levels, insulin resistance, and leptin concentrations while increasing adiponectin contents; however, these findings have to be interpreted cautiously and implemented under appropriate medical supervision (Albosta and Bakke, 2021).

In summary, the available evidence suggests that low-carbohydrate diets and intermittent fasting may be useful for achieving relatively rapid short-term metabolic improvements, whereas Mediterranean, anti-inflammatory, and low-glycemic dietary patterns may offer more sustainable benefits for long-term cardiometabolic health. Based on these observations, a phased, individualized strategy may serve as an evidence-informed consideration rather than definitive guidance: short-term intensive dietary strategies might be discussed for individuals where rapid glycemic or weight reduction is prioritized, followed by a transition to nutrient-dense, culturally appropriate, and sustainable dietary patterns for long-term diabetes prevention.

### Evidence-informed dietary considerations based on meal pattern models

Based on the synthesis of ten RCTs by this review, dietary management for prediabetes should not rely on a single, uniform prescription. Instead, dietary strategies must be individualized per metabolic phenotype, cardiometabolic risk profile, cultural food context, and likelihood of long-term adherence. However, the current evidence has to be interpreted with caution. The included studies were characterized by substantial heterogeneity among study populations, intervention components, durations, and outcome measures. Moreover, the overall certainty of evidence was limited by methodological concerns, with 50% judged to have a high risk-of-bias, 40% raising some concerns, and only 10% assessed as having a low risk. Within this context, four major dietary intervention models and their plausible biological mechanisms are discussed below.

***Calorie- and carbohydrate-restricted diets*:** Interventions emphasizing total energy restriction and reduced carbohydrate intake, including Diets 1, 6, and 12, produced relatively rapid short-term improvements in glycemic control. For example, Röhling et al. (2020) reported that a low-carbohydrate formula-based meal replacement strategy enabled 50% of participants to achieve normoglycemia after 52 weeks. These findings suggest that structured calorie- and carbohydrate-restricted diets may be particularly useful for individuals requiring early metabolic improvement, especially when obesity and insulin resistance are prominent features.

The primary biological mechanism underlying these impacts is likely the induction of a sustained energy deficit. Caloric restriction promotes clinically meaningful weight loss, commonly in the range of 7% – 10% of baseline body weight, thereby improving peripheral insulin sensitivity and reducing the secretory burden on pancreatic β-cells (Velázquez-López et al., 2013; Evert et al., 2019). In addition, the relatively elevated protein contents of these interventions, often accounting for 15% – 25% of the total energy intake, may enhance endogenous incretin responses, particularly glucagon-like peptide-1 secretion, increase diet-induced thermogenesis, and promote satiety. These impacts may also help preserve lean body mass during weight loss, a consideration crucial for individuals at risk of sarcopenic obesity or metabolically unfavorable weight reduction (Tettamanzi et al., 2021; Flores-Hernández et al., 2024; Reynolds and Mitri, 2024).

Despite these promising metabolic effects, the strength of evidence for this dietary model remains limited. Strict carbohydrate or calorie restriction is often difficult to sustain over time, and declining adherence may attenuate its effectiveness in real-world settings. Furthermore, as dietary interventions cannot be fully blinded, performance bias remains an inherent methodological concern in trials evaluating these approaches (Velázquez-López et al., 2013; Röhling et al., 2020; Buso et al., 2021). Therefore, calorie- and carbohydrate-restricted diets may be most appropriately considered as structured, time-limited, and closely monitored strategies rather than universal long-term prescriptions for all individuals with prediabetes (Velázquez-López et al., 2013; Lean et al., 2018; Chair et al., 2022).

***Carbohydrate-quality-focused approaches*:** Dietary interventions prioritizing complex carbohydrates and low-glycemic food sources, including Diets 8, 9, 11, 13, and 14, suggest that carbohydrate quality may have clinical relevance beyond simply reducing carbohydrate quantity. Replacing refined foods such as white rice with whole grains, legumes, and other minimally processed carbohydrate sources, as observed in Diet 11, improved intermediate metabolic markers, including HbA1c and HOMA-IR, even in the absence of substantial weight loss (Wolever and Mehling, 2002, 2003; Kim et al., 2014; Lean et al., 2018; Kazemi et al., 2021).

Several biological mechanisms may explain these benefits. Low-glycemic-index foods attenuate postprandial glu-

cose oscillations and reduce postprandial insulin demand. Over time, this effect may improve β-cell compensatory capacity, reflected by a more favorable disposition index, and support metabolic adaptation during peripheral insulin resistance (Wolever and Mehling, 2002, 2003; Kim et al., 2014). In parallel, soluble and viscous fibers from whole grains and legumes are fermented by the gut microbiota, producing short-chain fatty acids, particularly propionate and butyrate. These metabolites may activate G-protein-coupled receptors, including GPR41 and GPR43, thereby enhancing β-cell function and insulin signaling, but reducing low-grade systemic inflammation (Evert et al., 2019; Zhang et al., 2022; Iizuka and Yabe, 2023).

Nevertheless, an interpretation of the influence of this intervention category requires caution. The included trials differed substantially in their definitions of "complex carbohydrates," the specific food substitutions used, the amount and type of fiber provided, and the intensity of background dietary counseling. Thus, carbohydrate-quality-focused interventions must be viewed less as acute reversal strategies and more as sustainable metabolic-support approaches. Their greatest applicability may lie in improving postprandial glycemic stability, enhancing overall diet quality, and supporting long-term cardiometabolic risk reduction (Wolever and Mehling, 2002; Kim et al., 2014, 2023; Albosta and Bakke, 2021).

***Mediterranean, Jiangnan, and anti-inflammatory dietary patterns*:** Whole-diet approaches based on culturally defined food patterns, including Diets 2, 3, 10, 15, 17 and 18, also demonstrated favorable metabolic potential. Mediterranean-style and traditional Jiangnan diets were associated with an average weight reduction of ∼7% after 6 months and appeared to support visceral fat reduction. In the anti-inflammatory dietary intervention evaluated by Pasdar et al. (2025), a notable trend toward lower hs-CRP levels suggested potential benefits for systemic inflammation (Wolever and Mehling, 2002, 2003; Qi and Hu, 2007; Luo et al., 2022).

The metabolic benefits of these dietary patterns are likely attributable to the overall food matrix rather than to any single nutrient. A key feature is improved lipid quality, particularly the replacement of saturated fatty acids with monounsaturated and polyunsaturated ones derived from olive oil, nuts, seeds, and fatty fish. Such a shift may enhance cell membrane fluidity, thereby supporting insulin receptor structure, insulin signaling, and glucose transport. In addition, plant-derived bioactive compounds, including green tea polyphenols, curcumin from turmeric, and anthocyanins from berries, may modulate glucose metabolism. These compounds may mildly inhibit α-amylase activity, slow starch digestion, and activate adenosine monophosphate-activated protein kinase, a pivotal cellular energy sensor involved in glucose uptake, lipid oxidation, and mitochondrial function (Louheranta et al., 2002; Yang et al., 2017; Mehta et al., 2021; Luo et al., 2022; Zin et al., 2022; American Diabetes Association Professional Practice Committee, 2024; González et al., 2024; Pasdar et al., 2025).

Evidence for these culturally embedded dietary patterns must be interpreted contextually. Mediterranean and Jiangnan dietary models are prominently shaped by regional food availability, culinary traditions, and habitual eating practices, which may limit their direct transferability across populations. Furthermore, many trials relied heavily on self-reported dietary records, increasing the possibility of recall bias and dietary misclassification. Therefore, although these whole-diet patterns appear promising, their real-world implementation should emphasize cultural adaptation, food accessibility, and behavioral sustainability rather than rigid replication of a specific regional diet (Huo et al., 2015; DeSalvo et al., 2016; Huntriss et al., 2018; Röhling et al., 2020; Luo et al., 2022).

***Intermittent fasting*:** Intermittent fasting strategies, including Diets 4, 5, 7, and 16, have emerged as potentially useful approaches for weight management and improvement of insulin dynamics in individuals with prediabetes. In the included evidence, ADF was associated with remarkable reductions in body weight and BMI, with weight loss reaching up to 14.7 kg among participants. TRE, particularly the 16:8 regimen, also favorably affected lipid profile stabilization, suggesting potential cardiometabolic benefits beyond weight reduction alone (Bartholomew et al., 2021; Chair et al., 2022; Pasdar et al., 2025).

The biological rationale for intermittent fasting centers on metabolic switching. During prolonged fasting, the body gradually shifts from glycogen-derived glucose utilization toward lipolysis and fatty acid oxidation, with ketone bodies serving as alternative energy substrates. Such a metabolic state may stimulate autophagy, improve mitochondrial efficiency, and enhance insulin sensitivity through mechanisms partly independent of weight loss. From the perspective of appetite regulation, fasting regimens may also influence peptide YY, a satiety-related hormone, and modify postintervention ghrelin dynamics. These hormonal adaptations may contribute to enhanced appetite control and support a lower bodyweight set point in some individuals (Bartholomew et al., 2021; Milenkovic et al., 2021; Chair et al., 2022; Pasdar et al., 2025).

Nevertheless, intermittent fasting should be regarded as a conditional rather than a universally applicable strategy; trials have reported relatively high dropout rates, raising concerns about acceptability and long-term adherence. In addition, robust long-term data from trials with a low risk-of-bias remain limited, particularly regarding safety among individuals with complex comorbidities, older adults, or those on glucose-lowering medications.

Therefore, intermittent fasting must be implemented with clinical supervision, careful screening, and individualized monitoring rather than being broadly recommended as a stand-alone intervention for all individuals with prediabetes (Albosta and Bakke, 2021; Huang et al., 2023; Sun et al., 2024).

Overall, the evidence available indicates that multiple dietary strategies can improve glycemic and cardiometabolic outcomes in individuals with prediabetes, but their mechanisms, feasibility, and sustainability vary substantially. Energy- and carbohydrate-restricted diets may be useful for achieving rapid metabolic improvement, whereas carbohydrate-quality-focused approaches and whole-diet patterns may offer more sustainable benefits. Intermittent fasting may be effective for specific individuals but requires careful clinical oversight. Future trials must use standardized dietary definitions, objective adherence measures, longer follow-up periods, and rigorous risk-of-bias control to better determine dietary patterns that are most effective, safe, and scalable for different subgroups of individuals with prediabetes (Velázquez-López et al., 2013; Röhling et al., 2020; Buso et al., 2021; Chair et al., 2022; Luo et al., 2022).

### Influence of exercise

As this review focused on dietary interventions, physical activity was generally treated as a control variable or supportive lifestyle component rather than as an independent intervention. In most included RCTs, participants were instructed to maintain their usual activity levels, while physical activity was commonly assessed via self-reported questionnaires and adjusted for statistical analyses (Velázquez-López et al., 2013; Buso et al., 2021; Chair et al., 2022).

Some studies provided generalized exercise advice, such as encouraging walking, jogging, or increased movement; however, exercise prescriptions were either not standardized, not analyzed separately, or had low adherence. For example, in the study by Velázquez-López et al. (2013), only ~10% of participants adhered to the recommended aerobic exercise regimen, suggesting that the improvements observed were primarily attributable to dietary modifications (Velázquez-López et al., 2013; Pasdar et al., 2025).

Overall, although residual confounding from physical activity cannot be fully excluded, the metabolic benefits reported here are likely to be driven predominantly by dietary interventions. Future studies should incorporate objective physical activity measurements to better distinguish the independent and synergistic effects of diet and exercise.

### Strengths and limitations

This review has several strengths; it provided an up-to-date synthesis of evidence till 2025, aligned with current clinical guidance (e.g., ADA 2026), and included only RCTs, ensuring a high level of confidence. The inclusion of diverse dietary patterns (e.g., Mediterranean, low-carbohydrate, high-protein) offers a comprehensive overview of nutritional strategies applicable for controlling prediabetes. Methodological quality was rigorously appraised using the Cochrane RoB 2 tool, enhancing transparency.

Additionally, the limitations should be acknowledged; the small number of studies (n=10) and modest total sample size (n=1,098) may limit generalizability. A substantial proportion of trials were judged to have "some concerns" or "high risk" of bias, largely due to challenges in blinded dietary interventions. Restriction to publications only in English may have excluded culturally specific practices. In addition, most studies were short-term (12 – 52 weeks) and focused on metabolic endpoints, with limited data on long-term sustainability, adherence, and patient-centered outcomes (e.g., quality of life or psychological effects).

### Future directions

Future studies must prioritize large, multicenter RCTs with follow-up beyond 24 months to evaluate the durability of dietary effects on glycemic status and diabetes prevention. Greater attention should be given to adherence, feasibility, and real-world sustainability. They should also assess early vascular risk markers and long-term cardiometabolic trajectories in adults with prediabetes, without extrapolating from populations with established T2D. Further, direct comparisons between dietary models, standardized outcomes, and objective adherence measures are needed. Digital health tools, continuous glucose monitoring, and personalized nutrition approaches may support individualized intervention delivery. Patient-reported outcomes, including quality of life, dietary burden, and psychological well-being, must also be incorporated to ensure that interventions are clinically meaningful and sustainable.

## CONCLUSION

This systematic review suggests that structured dietary interventions may improve glycemic and select cardiometabolic outcomes in adults with prediabetes, although the effects observed varied across dietary patterns, metabolic targets, intervention duration, and feasibility of long-term adherence. Among the RCTs included, calorie- and carbohydrate-restricted approaches produced relatively rapid short-term improvements in body weight and glycemic outcomes, whereas low-glycemic and fiber-rich dietary approaches supported insulin sensitivity

and β-cell function with less dependence on weight reduction. Mediterranean, Jiangnan, and anti-inflammatory dietary patterns provided wider cardiometabolic benefits, while intermittent fasting may be a feasible option for certain individuals, if adherence, safety, and lipid responses are carefully considered.

Overall, no single dietary strategy emerged as universally optimal for all adults with prediabetes. Dietary approaches therefore need to be individualized per the metabolic profile, cardiovascular risk, cultural context, preferences, and long-term feasibility. Given the heterogeneity in the available evidence and the absence of a formal certainty of data assessment, these findings should be interpreted as potential dietary considerations rather than guideline-level recommendations. Future randomized trials should incorporate longer follow-ups, direct comparisons among dietary models, objective adherence measures, and patient-centered outcomes to better inform sustainable and scalable strategies for diabetes prevention.

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## AUTHOR DISCLOSURE STATEMENT

The authors declare no conflict of interest.

## AUTHOR CONTRIBUTIONS

Concept and design: HTN, TN. Analysis and interpretation: HTP, PDP. Data collection: HML, DTN, DKL. Writing the article: HTN, TN. Critical revision of the article: HTP, PDP, HML, DTN, DKL, NBN, GHH. Final approval of the article: All authors. Statistical analysis: NBN, HTP. Overall responsibility: GHH.

## SUPPLEMENTARY MATERIALS

Supplementary materials can be found via https://doi.org/10.3746/pnf.2026.104

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